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PEAS Description and Properties of Task Environments

PEAS stands for Performance measure, Environment, Actuators and Sensors. Each task environment is also classified as fully/partially observable, single/multi agent, deterministic/stochastic, episodic/sequential, static/dynamic, discrete/continuous and known/unknown.

A) Playing soccer

PEAS Description

Performance measure: Goals scored, winning the match, team coordination, fair play (few fouls).

Environment: Soccer field, ball, goals, teammates, opposing players, referee, crowd, weather.

Actuators: Legs/feet (run, kick, pass, dribble), head (header), body (tackle, block), voice/gestures (communicate).

Sensors: Eyes/camera (ball, players, field lines), ears (calls, whistle), touch and balance sensors, position and speed sensors.

Properties of the Environment

Observable: Partially – Cannot see players behind or opponents' intentions; ball and players are only partly visible.

Agents: Multi-agent – Cooperative with teammates and competitive against opponents.

Determinism: Stochastic – Ball bounce, opponents' moves and weather are unpredictable.

Episodic: Sequential – Current actions affect the rest of the match.

Static: Dynamic – Players and ball keep moving while the agent thinks.

Discrete: Continuous – Positions, speeds and time are continuous.

Known: Known – Rules and physics of the game are known.

B) Exploring the subsurface oceans of Titan

PEAS Description

Performance measure: Area/volume explored, scientific data and samples collected (chemistry, signs of life), probe survival, energy efficiency, data transmitted to Earth.

Environment: Titan's subsurface liquid ocean, ice crust, very low temperature, high pressure, unknown terrain, communication link to Earth.

Actuators: Thrusters/propellers and rudders (movement), drill or melt probe, sampling arm, instruments control, radio transmitter.

Sensors: Sonar, cameras, temperature, pressure and chemical sensors, gyroscope/IMU, depth gauge, radio receiver.

Properties of the Environment

Observable: Partially – Dark, murky ocean; sensors give only local information.

Agents: Single agent – No other agents in the environment (unless unknown life exists).

Determinism: Stochastic – Currents, ice movement and equipment failures are uncertain.

Episodic: Sequential – Each move affects battery, position and later exploration.

Static: Dynamic – Currents and the environment change continuously.

Discrete: Continuous – Depth, position, temperature, pressure are continuous.

Known: Unknown – Environment is largely unexplored; the agent must learn it.

C) Shopping for used AI books on the Internet

PEAS Description

Performance measure: Low price, good book condition, relevance to AI topics, seller reliability, fast delivery, staying within budget.

Environment: Internet, online bookstores/marketplaces, search engines, sellers, reviews, payment and shipping systems.

Actuators: Typing search queries, clicking links, filtering/sorting, comparing, adding to cart, filling forms, making payment.

Sensors: Web page content (titles, prices, conditions, ratings, reviews), search results, order confirmations.

Properties of the Environment

Observable: Partially – Only the pages visited are seen; stock, true condition and other offers are hidden.

Agents: Single agent – Other buyers and sellers exist, but can be treated as part of the environment (multi-agent if bidding).

Determinism: Stochastic – Stock, prices and delivery can change unpredictably.

Episodic: Sequential – Earlier searches and choices affect later purchases and budget.

Static: Dynamic – Prices and availability change while browsing.

Discrete: Discrete – Finite pages, links, books and actions.

Known: Known – How websites, carts and payments work is known.

D) Playing a tennis match

PEAS Description

Performance measure: Winning points, games and sets; accuracy and power of shots; few unforced errors; endurance.

Environment: Tennis court, net, ball, rackets, opponent, umpire/line judges, weather and surface.

Actuators: Legs (move), arm and racket (forehand, backhand, serve, volley), body rotation, voice (call the score).

Sensors: Vision (ball, opponent, lines), hearing, touch/grip feedback, balance and body-position sensors.

Properties of the Environment

Observable: Partially – Ball spin and opponent's intention are not fully visible.

Agents: Multi-agent – Competitive (opponent); cooperative in doubles.

Determinism: Stochastic – Bounces, wind and opponent's shots are uncertain.

Episodic: Sequential – Each shot affects the following rally and score.

Static: Dynamic – The ball and opponent move while the agent decides.

Discrete: Continuous – Ball position, speed and spin are continuous.

Known: Known – Game rules and physics are known.

E) Practicing tennis against a wall

PEAS Description

Performance measure: Number of consecutive returns, accuracy of shots, consistency, improved technique.

Environment: Wall, ground surface, ball, racket, weather conditions.

Actuators: Legs (move), arm and racket (swing and hit), body positioning.

Sensors: Vision (ball and wall), hearing (impact), touch/grip feedback, balance sensors.

Properties of the Environment

Observable: Partially – Ball spin and exact bounce are not fully known in advance.

Agents: Single agent – Wall is a passive object, not an agent.

Determinism: Stochastic – Slight randomness from wall surface, spin and wind (nearly deterministic).

Episodic: Sequential – Each return decides the next ball position.

Static: Dynamic – The ball keeps moving while the agent acts.

Discrete: Continuous – Ball motion and positions are continuous.

Known: Known – Physics of rebound is known.

F) Performing a high jump

PEAS Description

Performance measure: Height of the bar cleared, successful attempts, correct technique, safe landing.

Environment: Run-up area, take-off spot, bar and uprights, landing mat, wind, judges.

Actuators: Legs (run and take-off), arms (swing), body (arch/Fosbury flop), landing movements.

Sensors: Vision (bar and mat), balance, proprioception (body position), touch of foot on ground, timing sense.

Properties of the Environment

Observable: Partially – Wind and exact body position in mid-air are not fully perceivable.

Agents: Single agent – Only the jumper acts; judges do not influence the jump.

Determinism: Stochastic – Wind, take-off error and fatigue add uncertainty.

Episodic: Sequential – Within a jump, run-up, take-off and clearance depend on each other (separate attempts are nearly episodic).

Static: Semi-dynamic – Bar is static, but the jumper's own motion changes quickly; time matters.

Discrete: Continuous – Speed, angle, height and time are continuous.

Known: Known – Rules and physics are known.

G) Knitting a sweater

PEAS Description

Performance measure: Correct size and pattern, uniform stitches, no dropped stitches, finished sweater, time and yarn used.

Environment: Yarn, knitting needles, pattern/instructions, work surface.

Actuators: Fingers and hands (loop, pull, stitch, purl), needle movement, cutting/joining yarn.

Sensors: Vision (stitches, pattern), touch (yarn tension).

Properties of the Environment

Observable: Fully – All stitches and the pattern are visible to the knitter.

Agents: Single agent – Only the knitter acts.

Determinism: Deterministic – Next state is determined by the current state and action (apart from rare slips).

Episodic: Sequential – Each stitch affects the shape of the final sweater.

Static: Static – Nothing changes unless the agent acts.

Discrete: Discrete – Stitches, rows and patterns are countable.

Known: Known – Knitting rules and results are known.

H) Bidding on an item at an auction

PEAS Description

Performance measure: Winning the item at the lowest possible price, not exceeding the budget/valuation, maximising utility (value minus price).

Environment: Auction house or website, item, auctioneer, other bidders, bidding rules, current price and time.

Actuators: Place a bid, raise a bid, hold, withdraw, stop bidding.

Sensors: Current highest bid, other bidders' bids, auctioneer's announcements, remaining time, item description.

Properties of the Environment

Observable: Partially – Other bidders' maximum budgets and valuations are hidden.

Agents: Multi-agent – Competitive (other bidders compete for the same item).

Determinism: Stochastic – Other bidders' actions are unpredictable.

Episodic: Sequential – Each bid affects later bids and the final price.

Static: Dynamic – Bids and prices change while the agent deliberates.

Discrete: Discrete – Finite bid amounts and actions; discrete time steps.

Known: Known – Auction rules are known.

Summary

Activity	Observable	Agents	Determinism	Episodic / Sequential	Static / Dynamic	Discrete / Continuous	Known / Unknown
A) Playing soccer	Partially	Multi-agent	Stochastic	Sequential	Dynamic	Continuous	Known
B) Exploring the subsurface oceans of Titan	Partially	Single agent	Stochastic	Sequential	Dynamic	Continuous	Unknown
C) Shopping for used AI books on the Internet	Partially	Single agent	Stochastic	Sequential	Dynamic	Discrete	Known
D) Playing a tennis match	Partially	Multi-agent	Stochastic	Sequential	Dynamic	Continuous	Known
E) Practicing tennis against a wall	Partially	Single agent	Stochastic	Sequential	Dynamic	Continuous	Known
F) Performing a high jump	Partially	Single agent	Stochastic	Sequential	Semi-dynamic	Continuous	Known
G) Knitting a sweater	Fully	Single agent	Deterministic	Sequential	Static	Discrete	Known
H) Bidding on an item at an auction	Partially	Multi-agent	Stochastic	Sequential	Dynamic	Discrete	Known